

Precise Modeling and Analysis of DQ-Frame Current Controller for High Power Converters with Low Pulse Ratio

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Abstract—This paper presents a modeling and analysis methodology for the current loop control design for three-phase high power converters with extremely low sampling ratios. It applies the complex transfer function and the double-sided Frequency-Response-Functions to quantitatively characterize the system such as the phase margin, dynamic performance and the noise immunity. Both standard PI control and PI control with a current state feedback are modeled and validated via circuit simulations in SABER, which demonstrate the precision and the intuitiveness of the proposed modeling concept. Based on these precise models, some special phenomena and design challenges of the current control loops for high power converters are summarized followed by some generalized design recommendations. Since the precision of the modeling concept is independent of the sampling ratio, the methodology can also be leveraged to research topics such as the “negative impedance” and “active damping”. Generally, this simple and precise modeling concept is highly attractive for control loop analysis and design of three-phase converters.

Index Terms— Low sampling ratio, Complex transfer function modeling, double-sided Frequency-Response-Function, Decoupling

I. INTRODUCTION

High power converters have been widely applied for e.g. wind, photovoltaic and large electrical drive applications in the last decades. At such power and voltage level, the switching frequency of the converters is strongly limited with a typical range from several kilo Hertz (e.g. when using 1.2 kV or 1.7 kV IGBT modules [1]) down to only several hundred Hertz (when using e.g. 4.5 kV IGCT or IGBT press-packs [2]). At such sampling-to-fundamental low frequency ratios, the dynamic performance of the innermost control loop, i.e. the current controller, becomes a challenge.

A number of papers have focused on improving the performance of the current controllers. The well-known industrialized algorithms are synchronous frame Proportional-Integral (PI) controls, stationary frame Proportional-Resonant (PR) controls, and PI with state feedbacks [3] – [5]. Variable algorithms have been proven to

work well in real applications, as far as a sufficient sampling ratio is guaranteed.

On the other hand, it seems that the precise modeling of the control loop itself did not draw much attention in the past: most papers focus directly on the pole-zero placements with a rough control loop model, where the sampling effects are not fully explored and included [6] [7]. For low power converters using fast switches, controllers have sufficient bandwidth anyway (i.e. quasi-perfect controllers), and such approximations (or actually modeling errors) can be tolerated. However, for high power converters with very limited switching frequencies (e.g. 450 Hz carrier / 900 Hz sampling frequency with 50 Hz fundamental), such modeling errors become critical, and may lead to several serious issues, e.g.:

- The controller gain calculated from the simplified model “should” guarantee 45 ° phase margin but in reality the controller is unstable. Therefore users have to further reduce the controller gain manually.
- The controller is stable with reduced gain but has low dynamic performance. Therefore the converter power (current) capability has to be strongly de-rated to avoid a high current overshoot that exceeds the device’s SOA (Safe-Operating-Area).
- Voltage or current noises in the control loop may be significantly amplified at certain frequencies, which cannot be explained with the simplified models.
- Performance degradation: e.g. the d and q axis should be well-decoupled using the state feedback, but in reality a significant cross-coupling can still be observed.

Typically, when the modeling is precise, the controller configuration is not that critical. In contrast, when the control algorithm is developed based on an inaccurate model, its practical value is probably reduced as discussed above. Hence, a comprehensive understanding (i.e. a precise modeling) of the control loop is an essential precondition for a proper control design – especially for high power converters with extremely low sampling ratios.

A typical synchronous dq-frame PI control block diagram is illustrated in Fig. 1, which is a Multiple-Input-Multiple-Output system. If the controller has sufficient sampling ratios, it can be simplified as quasi a buck converter that is a Single-Input-Single-Output system, as shown in Fig. 2. However, since several control blocks such as phase compensations and abc-dq-abc transforms are excluded, the accuracy of the model is strongly degraded at low sampling ratios.

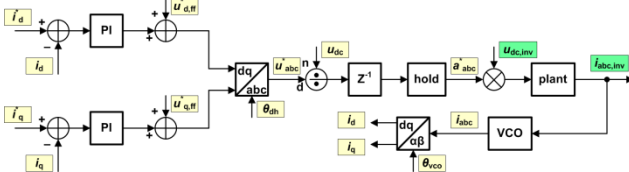


Fig. 1 Control Block Diagram of a dq-frame PI control with MIMO

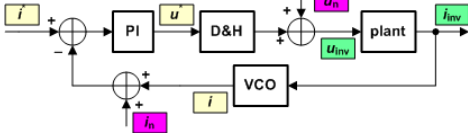


Fig. 2 Simplified single-phase PI Control with lost information

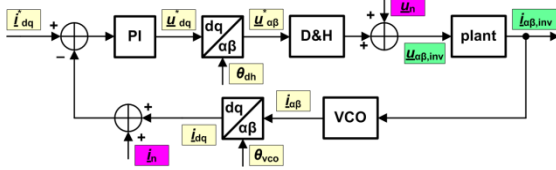


Fig. 3 CBD of PI control using complex transfer function

Modeling using matrix methods has been discussed in several papers [8] – [11], which can precisely describe the MIMO control loop. On the other hand, this MIMO based modeling seems to be relatively complicated and not intuitive. Sometimes the positive and negative sequences need to be analyzed separately.

Another modeling concept is to merge the d and q (or α and β) axis variables as a complex variable [12] – [16], so that the intrinsic MIMO system is simplified to a SISO, see Fig. 3. Using this approach, the explanations of the non-linear effects such as the frequency shift due to the $\alpha\beta$ -dq transforms become intuitive. Furthermore, by expanding the frequency axis to the negative side, the Frequency-Response-Function (FRF) of the control loop can be continuously plotted over a double-sided frequency range, which is similar to the Bode-diagram that power electronics engineers are familiar with.

Unfortunately, it seems that the superiority of the complex transfer function modeling concept has not been well recognized in the past. One potential reason is that the previous works did not focus on the extremely low sampling frequency applications. As discussed, at higher sampling ratios, the lost precision is not important, and the sampling effects are frequently excluded [6] [7]. Secondly, most papers use tools such as Root Locus [6] and Nyquist criterion [15] to design their control loop, but did not explain how to use these tools to precisely quantify the

characteristics of the system such as the phase margin, the bandwidth and the noise immunity. Actually, from this point of view, the FRF (e.g. the Bode Diagram) is an attractive analysis tool for power electronics engineers, if the modeling of the control loop is precise. However, the authors did not find any previous work about such modeling and analysis methodology (i.e. precise modeling using complex transfer function & quantitative analysis using FRF & generalized design criteria & rules discussions) that are valuable for three-phase current controller designs.

This paper attempts to cover this gap: a precise and simple modeling and analysis methodology for the current control loop is presented in this paper, which is initially explicitly developed for high power converters with extremely low sampling ratios. Firstly, the complex transfer function modeling concept is refreshed in Section II with a modeling example for a standard dq-frame PI control. The superiority of this modeling approach in terms of the modeling precision, simplicity and intuition is demonstrated via simulation validations in SABER (Section III). Afterwards, some interesting phenomena and design challenges of high power converter applications (with extremely low sampling ratios) are discussed in Section IV. Moreover, generalized analysis methodologies and design instructions for the control loop using such a complex transfer function based SISO system are also discussed. Finally, conclusions are given.

II. REVIEW OF COMPLEX TRANSFER FUNCTION MODELING CONCEPT

A. Negative sequence vs. “negative” frequency

In a symmetric three-phase system, the phase waveform patterns are identical but phase-shifted by 120° . It is known that both the 2nd and 4th harmonics in the phase signals are shifted to the 3rd harmonic after the abc-dq transform, but rotating in opposite directions, i.e. the 2nd harmonic is negative sequence and the 4th is positive [8]. The abc-dq transform shifts the positive sequent components e.g. the 4th and 7th to the “left”, and the negative sequent components e.g. the 2nd and 5th to the “right”, see Fig. 4 (a). A separate analysis for the positive and the negative sequence is done that finally achieves a high modeling precision, however, at the expense of an increased modeling complexity [8].

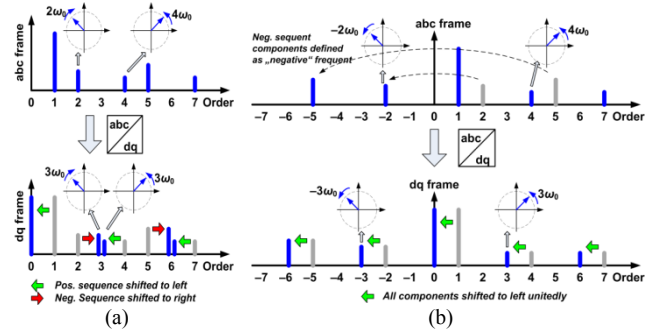


Fig. 4 Explanation of abc-dq transform using: (a) single-sided spectrum; (b) “negative” frequency concept

Interestingly, another explanation was also referred in [8]: if the 2nd harmonic is defined as a negative frequency component (i.e. -2^{nd}), the function of the abc-dq transform can be perfectly explained: it shifts the double-sided spectrum to the “left” by one fundamental, e.g. the -2^{nd} is shifted to -3^{rd} in the dq-frame, see Fig. 4 (b). Using this assumption, the modeling of the abc-dq transform becomes intuitive, since no separate modeling for the positive and negative sequence is necessary. However, the harmonics of a scalar variable are always positive. For three-phase systems, an additional concept – “sequence” is introduced. The next question is, what is the physical meaning of the “negative” frequency, i.e. does it represent the negative sequence of a three-phase system?

B. Complex vs. scalar Fourier Transform

It is well-known that the output spectrum of a complex Fourier Transform is double-sided (i.e. containing “negative” frequencies), since it assumes the input as a complex signal [17]. Even when the signal is “occasionally” scalar, the spectrum distribution is symmetrical around the dc component and the harmonics are conjugated, see Fig. 5. The conjugated components can be recognized as two harmonic components rotating at the identical speed but in opposite directions. Their imaginary parts are compensated and the real parts are added (i.e. doubled). Since a scalar signal intrinsically does not contain the “sequence” information, its negative frequency harmonics are typically flipped to the positive side, i.e. simplified to a classic single-sided spectrum that power electronics engineers are familiar with. However, if the signal has a complex form, its projections on both axes are fully defined by the real and imaginary part. Hence, it does contain the “sequence” information that can be expressed as the sign of each harmonic using a complex Fourier Transform (FT). More systematic explanations can be found in communication theory [17].

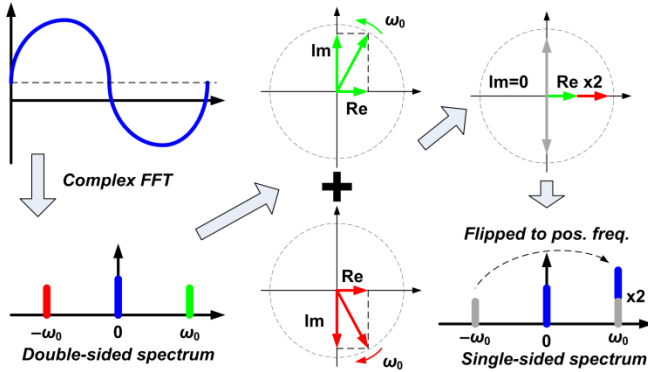


Fig. 5 Relationship between a complex FT and a scalar FT

C. Complex transfer function modeling for current loop

On the other hand, it is known that a three-phase abc-system can be expressed in the orthogonal $\alpha\beta 0$ -frame. For three-wire systems, the zero-sequence component is typically dropped since it does not generate currents, i.e. the system order is reduced to two ($\alpha\beta$ -frame). Some papers propose to describe the α and β axis variables in a complex form $\underline{x}(t) = x_\alpha(t) + jx_\beta(t)$ [12] – [14], so that the three-phase MIMO

system can then be simplified as a single-phase system with SISO. By extending the frequency to include the negative part [6] [7] [15] [16], this system is fully characterized and no separate analysis for the positive and negative sequence section is needed anymore. This intuitive expression ensures a continuous FRF over a double-sided frequency axis (from the negative to positive infinite), which is very similar to the Bode diagram that the power electronics engineers are familiar with.

D. Example: precise modeling of a dq-frame PI controller

As a show-case, the precise modeling of a standard dq-frame PI control loop is demonstrated here via a complex transfer function modeling. The control block diagram has been presented in Fig. 3. It is noted that the noises $\underline{u}_n(t)$ and $\underline{i}_n(t)$ are also complex variables, i.e. they may contain negative sequent noises.

1) Frame transformation

As explained in Fig. 4 (b), the abc($\alpha\beta$)-dq transform shifts the FRF to the “left” by ω_0 . Similarly, the dq- $\alpha\beta$ inverse transform shifts the FRF pattern to the “right” by ω_0 .

$$s_{\alpha\beta} = s_{dq} + j\omega_0 \quad (1)$$

Notice that the subscripts of the Laplace term s indicate the reference frames, e.g. $s_{\alpha\beta}$ refers to the $\alpha\beta$ -frame. For example, an L-load has a pole at dc when observed from the $\alpha\beta$ -frame. So if observed from the dq-frame, the plant pole is shifted to $-\omega_0$, see Fig. 6 (a).

$$\text{Plant} = \frac{1}{s_{\alpha\beta} L} = \frac{1}{(s_{dq} + j\omega_0) L} \quad (2)$$

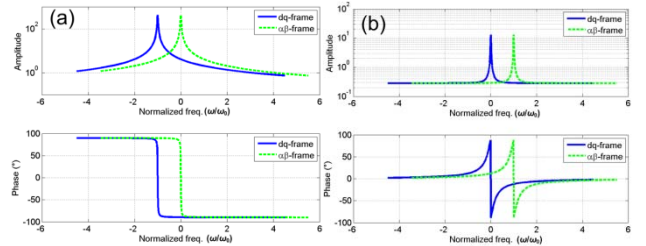


Fig. 6 FRF observed from dq and $\alpha\beta$ -frame, (a): an L-load; (b): PI regulator

Similarly, the PI pole has a pole at dc in the dq-frame, but is shifted to $+\omega_0$ if observed from the $\alpha\beta$ -frame, see Fig. 6 (b).

$$\text{PI} = K_p + \frac{K_I \cdot T_{sa}}{1 - e^{-s_{dq} \cdot T_{sa}}} = K_p + \frac{K_I \cdot T_{sa}}{1 - e^{-(s_{\alpha\beta} - j\omega_0) \cdot T_{sa}}} \quad (3)$$

It is noted that the z term is precisely described in the s -domain as $e^{sT_{sa}}$. Approximations such as the Tustin-Transformation are not necessary.

2) Delays and phase compensation

After the dq- $\alpha\beta$ transform block, there is a 1.5 sampling period delay, where “1” is caused by the computational delay

and “0.5” by the “hold” [8]. This leads to a phase drop that is proportional to the frequency, see the dashed-black line in Fig. 7. Typically, a phase compensation is applied during the dq- $\alpha\beta$ transform to shift this D&H phase curve to the top ($\theta_{dh}=1.5\theta_{sa}$, see Fig. 3), so that the phase drop is compensated at the dc (0 Hz) in the dq-frame, see the real-blue line. Equivalently, it compensates the phase drop at ω_0 in the $\alpha\beta$ -frame (see the dashed-green line).

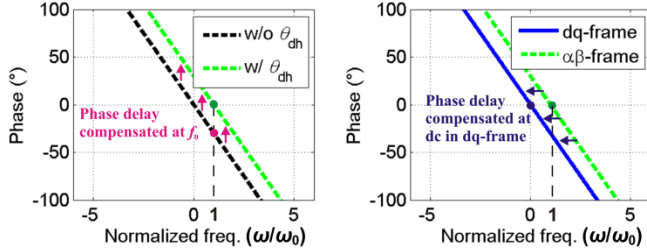


Fig. 7 FRF of Delay & Hold and impact of $1.5T_{sa}$ phase compensation

Combing the delay, hold and the phase compensation term (marked as subscript w/FF), its FRF is:

$$\begin{aligned} DH_{w/FF} &= \text{si} \left(\pi \frac{\omega_{\alpha\beta}}{\omega_{sa}} \right) e^{-1.5s_{\alpha\beta}T_{sa}} \cdot e^{j1.5\omega_0T_{sa}} \\ &= \text{si} \left(\pi \frac{\omega_{dq} + \omega_0}{\omega_{sa}} \right) e^{-1.5s_{dq}T_{sa}} \end{aligned} \quad (4)$$

It can be observed that the phase compensation is just a complex constant. It generates a phase bias to compensate the phase drop caused by the delay effect only at a targeted frequency, i.e. at dc in the dq-frame or at ω_0 in the $\alpha\beta$ -frame. In other words, the phase compensation can shift the phase FRF curve, but does not modify the phase slope of the delay terms.

It should be noted that the “hold” term generates not only $0.5T_{sa}$ phase delay, but also an amplitude distortion. On the other hand, even at 9 times carrier ratio (leading to 18 times sampling ratio), this amplitude distortion is negligible and therefore not plotted in Fig. 7.

When a Voltage-Controlled-Oscillator is applied (VCO, which is similar to a delta-sigma AD converter [18]), additional amplitude distortion and phase delay are generated in the control loop. Accordingly, a phase compensation term is activated (with $\theta_{vco}=0.5\theta_{sa}$) during the $\alpha\beta$ -dq transform:

$$\begin{aligned} VCO_{w/FF} &= \frac{1 - e^{-s_{\alpha\beta}T_{sa}}}{s_{\alpha\beta}T_{sa}} \cdot e^{j0.5\omega_0T_{sa}} \\ &= \frac{1 - e^{-(s_{dq} + j\omega_0)T_{sa}}}{(s_{dq} + j\omega_0)T_{sa}} \cdot e^{j0.5\omega_0T_{sa}} \end{aligned} \quad (5)$$

3) Open-loop transfer function

By simply combining these elements (1) – (5), the open-loop FRF of Fig. 3 is obtained:

$$G_{PI,OL}(s) = PI \cdot DH_{w/FF} \cdot \text{Plant} \cdot VCO_{w/FF} \quad (6)$$

This FRF of the open-loop transfer function $G_{PI,OL}(s)$ is plotted in Fig. 8 by replacing the Laplace term s_{dq} with $j\omega_{dq}$, or replacing the $s_{\alpha\beta}$ with $j\omega_{\alpha\beta}$. Interestingly, this double-sided transfer function contains two poles (at $-\omega_0$ and dc), which are not symmetrical around the y-axis. It should be noted that with the simplified modeling (Fig. 2) where the frequency shifts ($\alpha\beta$ -dq- $\alpha\beta$) are ignored, these two poles are overlapped at dc in the dq-frame, see the red dashed FRF curves in Fig. 13. Their accuracy will be compared in the next section.

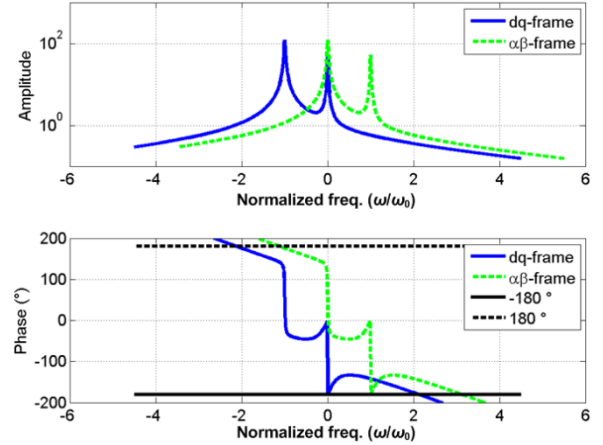


Fig. 8 FRF of Open-loop TF of a dq-frame PI controller with an L-load

III. VALIDATIONS VIA CIRCUIT SIMULATIONS

A. Setup for the circuit simulations

A three-level NPC converter with a digital control unit is generated in the circuit simulator SABER. The carrier frequency is 9 times the fundamental, where the actual sampling ratio is doubled due to the dual-edge sampling (18 times). Three inductors (0.15 per-unit impedance) in series with fundamental-frequency voltage sources are applied as an L-load. For medium-voltage applications, the series resistance of the inductor can be neglected.

B. “Original” PI control setup

Initially, the PI parameters are calculated according to [8], where a simplified model is applied (see Fig. 2). The K_p and K_i are applied to the digital dq-frame PI current controller (see Fig. 1) in SABER, where 45° phase margin “should” be guaranteed at different PI gain setups.

1) Qualitative validation via open-loop

The first simulation run shows that the system is unstable with an aggressive PI corner frequency $\omega_c/\omega_{pi}=5$ (ω_c is the open-loop cross-frequency where the gain is unity [8])! Even by further reducing the PI corner frequency to $\omega_c/\omega_{pi}=20$ where the controller becomes marginally stable, significant oscillations of the dq-currents are observed in the step response, see Fig. 10. Apparently, the simplified model has poor accuracy, while the PI gain is not configured properly.

To understand this degraded performance of the controller, the PI parameters are imported to (6) and

compared to the simplified model, see Fig. 13. It is noted that the calculated phase margins from the double-sided FRF and from the simplified modeling are significantly different! At positive x-axis the “actual” phase margin is even higher (65.6° vs. 42.6°). However, at the negative frequency there is almost no phase margin left (3.95°). This explains why the real control loop is almost unstable.

2) Quantitative validation via closed-loop

As a quantitative comparison, the noise amplification of the control loop is characterized and compared to the simulations: A current noise is injected to the feedback dq-current with defined amplitude but variable frequencies (can be negative) in the SABER model (see the pink signal shown in Fig. 3). By measuring the noise amplification in the phase currents, a closed-loop transfer function $\hat{i}_{dq}/\hat{i}_n$ over the double-sided frequency range is obtained. Similar to (6), the analytical expression of the closed-loop transfer function $\hat{i}_{dq}/\hat{i}_n$ is deduced from Fig. 3.

The results are plotted in Fig. 16. It is noted that the noise immunity of the closed-loop is poor at the negative frequency $-2.16\omega_0$ with 18 times amplifications! This resonant frequency is consistent with the observations in Fig. 10. It can be seen that the analytical expression is overlapped with the SABER measurements at different frequencies. The precision of the modeling is well-demonstrated.

C. Optimized PI control with disabled phase compensation

The poor performance of the original controller setup is mainly due to the unbalanced phase margins at two ends (65.63° at positive and 3.95° at negative end, see Fig. 13). As a “quick-fix” solution, the phase compensation term for the dq- $\alpha\beta$ transform unit is now disabled with purpose, i.e. $\theta_{dh}=0$. This has no impact on the amplitude FRF of the open-loop transfer function, but shifts the phase FRF to the bottom as discussed in Section II.D.2). Using this setup, the phase margins at both ends are balanced (35.78° vs. 34.11°, see Fig. 14), and the closed-loop resonance gain at the negative end ($-2.16\omega_0$) is now reduced by 10 times (Fig. 17 vs. Fig. 16). Finally, the oscillations disappeared in the step response in SABER as expected (Fig. 11).

D. Original PI setups with state feedback

Comparing the phase FRF of the open-loop transfer functions in Fig. 13 and Fig. 14, it is noted that the phase margins at two ends cannot achieve 45° at the same time, because the total margin $\varphi_{m,tot}$ (sum of the positive and negative margin $\varphi_{m,tot}=\varphi_{m-}+\varphi_{m+}$) is not increased, see TABLE I. Strictly speaking, the optimized PI controller with disabled D&H phase compensation is not properly damped yet. To increase the total phase margin, the controller structure should be modified. A standard concept is to use a current state feedback (SFB) to recentralize the plant pole back to the dc [6], see the blue term in Fig. 9.

Similar to (5), the open and closed-loop transfer functions of the PI+SFB can be precisely described in a complex form. Comparing the open-loop FRF (Fig. 14 vs. Fig. 15), it is noted that the total phase margin using PI+SFB

is significantly increased from 69° to 109°. Meanwhile, the cross-frequency at the positive end ω_{c+} is intrinsically increased from $0.18\omega_0$ to $0.8\omega_0$. As a consequence, the dynamic performance of the controller is significantly improved, as shown in TABLE I and Fig. 12. Detailed explanations are given in Section IV.C.

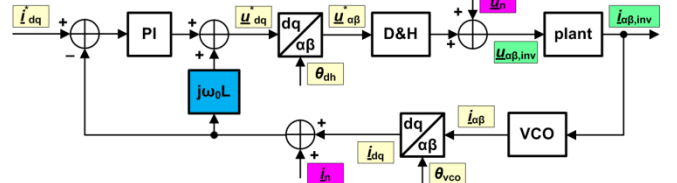


Fig. 9 PI with State feedback using complex transfer function

On the other hand, it is seen that the SFB leads to unbalanced phase margins, where the positive frequency end has less margin. This is due to the delays in the SFB loop that causes a phase drop, which can be analytically quantified using the proposed modeling method. Moreover, by adding additional phase compensation after the PI unit, the phase FRF of the open-loop TF can be shifted to equally distribute these two phase margins. Due to the limited space, a detailed and comprehensive analysis is planned for a separate paper.

TABLE I
PHASE MARGIN AND BANDWIDTH COMPARISON: PI VS. PI+SFB

Control setup	Original PI	Optimized PI	Original PI+SFB
φ_{m-} (neg.)	3.95°	34.11°	81.54°
φ_{m+} (pos.)	65.63°	35.78°	27.13°
$\varphi_{m,tot}$ (total)	69.58°	69.89°	108.67°
ω_{c-} (neg.)	$-2.12\omega_0$	$-2.12\omega_0$	$-1.17\omega_0$
ω_{c+} (pos.)	$0.18\omega_0$	$0.17\omega_0$	$0.8\omega_0$
$\omega_{c,tot}$ (total)	$2.3\omega_0$	$2.29\omega_0$	$1.97\omega_0$

IV. DISCUSSIONS

A. Phase margin balancing

When designing the current control loop using double-sided FRF, two phase margins on both ends have to be considered. For standard dq-frame PI controls, the FRF of the open-loop transfer function is intrinsically not symmetrical because the plant pole is shifted to $-\omega_0$. Standard phase compensation setup (i.e. $\theta_{dh}=1.5\theta_{sa}$) shifts the phase FRF to the top, which will reduce the margin at the negative end. Especially for high power converters with low sampling ratios, this setup may cause the PI controller to become unstable. In such cases, it is recommended to tune the compensation angle manually, e.g. as discussed in Section III.C.

In contrast, a typical SFB leads to a reduced phase margin at the positive end, which is due to the delays in the SFB loop. The phase compensation after the PI unit can be applied to balance the margins on two ends.

Hence, it can be concluded that for high power converters with extremely low sampling ratios, properly tuned phase compensations can shift the phase margin between the positive and negative ones to improve the stability of the control loop.

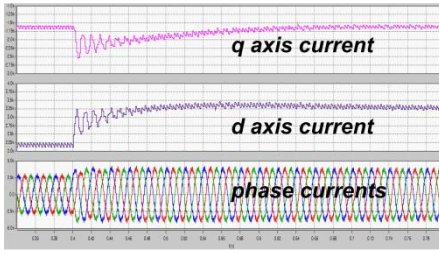


Fig. 10 Step response in SABER with original PI parameter setups: $\omega_c/\omega_{PI}=20$, $\theta_{dh}=1.5\theta_{sa}$, $\theta_{vco}=0.5\theta_{sa}$

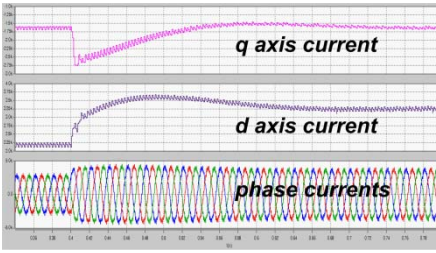


Fig. 11 Step response in SABER with modified PI parameter setups: $\omega_c/\omega_{PI}=20$, $\theta_{dh}=0$, $\theta_{vco}=0.5\theta_{sa}$

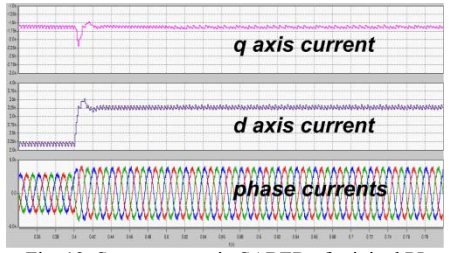


Fig. 12 Step response in SABER of original PI parameter setups + state feedback

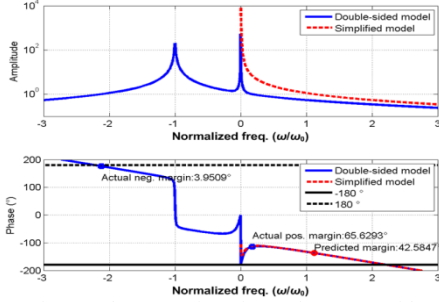


Fig. 13 Phase margins of open-loop FRF with $\omega_c/\omega_{PI}=20$, $\theta_{dh}=1.5\theta_{sa}$, $\theta_{vco}=0.5\theta_{sa}$

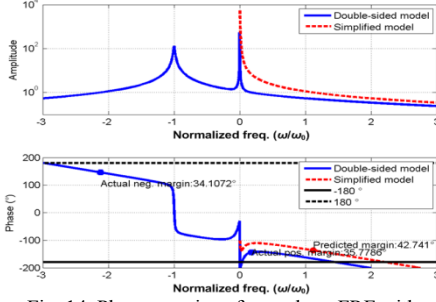


Fig. 14 Phase margins of open-loop FRF with $\omega_c/\omega_{PI}=20$, $\theta_{dh}=0$, $\theta_{vco}=0.5\theta_{sa}$

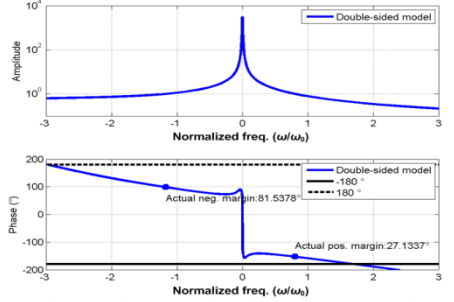


Fig. 15 Phase margins of open-loop FRF of PI+SFB, $\omega_c/\omega_{PI}=20$, $\theta_{dh}=1.5\theta_{sa}$, $\theta_{vco}=0.5\theta_{sa}$

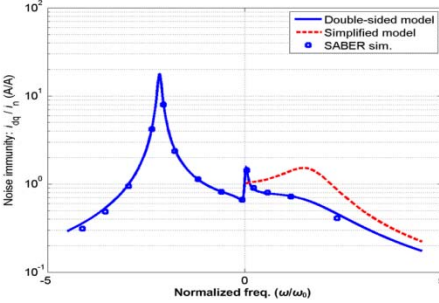


Fig. 16 Noise amplification: analytical model vs. SABER simulation with original PI parameter setups: $\omega_c/\omega_{PI}=20$, $\theta_{dh}=1.5\theta_{sa}$, $\theta_{vco}=0.5\theta_{sa}$

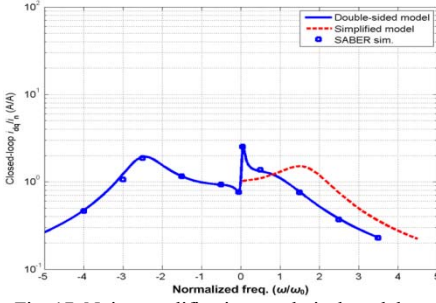


Fig. 17 Noise amplification: analytical model vs. SABER simulation with optimized PI parameter setups: $\omega_c/\omega_{PI}=20$, $\theta_{dh}=0$, $\theta_{vco}=0.5\theta_{sa}$

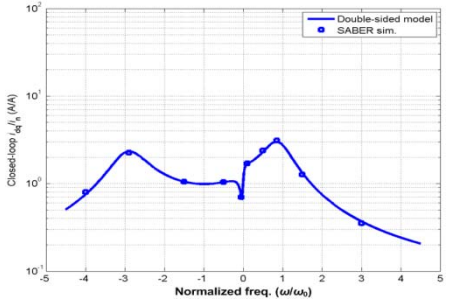


Fig. 18 Noise amplification: analytical model vs. SABER simulation with original PI parameter setups + SFB: $\omega_c/\omega_{PI}=20$, $\theta_{dh}=1.5\theta_{sa}$, $\theta_{vco}=0.5\theta_{sa}$

B. Total phase margin

Total phase margin is defined as the sum of the margins at the positive and negative ends, i.e. $\varphi_{m,tot}=\varphi_{m+}+\varphi_{m-}$. Only when it exceeds 90° , sufficient phase margins on both ends ($2 \times 45^\circ$) can be achieved by tuning the phase compensation as discussed above.

One straightforward method to increase the total margin is to reduce the cross frequencies on both ends by reducing the PI gain. Since the PI control loop contains two poles (one at $-\omega_0$ and one at dc with infinite gains), the cross frequencies of the amplitude FRF (where gain=1) are always located on the “left” side of the leftmost pole at $-\omega_0$ and on the “right” side of the rightmost pole at dc. From TABLE I it can be seen that the positive cross frequency using the PI control is about $0.18\omega_0$, which is already close to the dc. In this case, reducing the PI gain is not attractive, as it will further reduce the dynamic performance (details in section IV.C).

If the SFB is used as shown in TABLE I, the total cross frequency (defined as the $\omega_{c,tot}=\omega_{c+}+\omega_{c-}$) is reduced from $2.3\omega_0$ to $1.97\omega_0$. Accordingly, the total phase margin $\varphi_{m,tot}$ is

significantly increased from 70° to 109° . It should be noted that ω_{c-} is shifted to the “right” by about ω_0 compared to the PI control, since the plant pole is shifted to the dc and there is no constrain anymore that the left cross-frequency has to be located on the left side of $-\omega_0$. On the other end, the positive cross frequency is intrinsically increased from $0.18\omega_0$ to $0.8\omega_0$, i.e. the positive bandwidth is increased.

It can be concluded here that the PI+SFB can effectively increase the total phase margin without sacrificing the PI gain of the controller, which explains why the SFB is frequently applied for high power converter controls with extremely low sampling ratios.

C. Dynamic performance

Comparing the step response in Fig. 10, Fig. 11 and Fig. 12, it is found that the PI+SFB achieves a much higher dynamic performance compared to the PI controls, even without modifying the PI gain. Comparing the positive and negative cross frequencies in TABLE I, it can be found that the PI+SFB has much higher ω_{c+} , but lower ω_{c-} and also lower $\omega_{c,tot}$ than using PI. In this paper, the cross frequencies are defined as the bandwidths of the controller since it

contains the gain as well as the phase margin information. The question arising is: which one of the three parameters determines the dynamic performance of the controller?

The answer is visualized in Fig. 19: as discussed in Section II.B, the spectrum of a step in the complex system is double-sided, i.e. the 50 % of the energy is located at the negative frequencies (b). A fictive transfer function $G(s)$ is composed of two Low-Pass-Filters with 10 times bandwidth difference (2 Hz vs. 20 Hz, see (c)). The response curve contributed by the positive half $G(s)$ achieves a high dynamic (e) due to the high bandwidth. However, the response curve in (d) contributed by the negative half $G(s)$ is much slower. Notice that at steady-state each response curve contributes 50 % of the reference amplitude, and finally the observed response is the sum of (d) and (e). From (f) it is noted that the response curve initially has a high dynamic, which is contributed by the high bandwidth of the positive $G(s)$. However, the response curve is afterwards slowed down since it is then dominated by the negative bandwidth.

From this example it can be concluded that although a double-sided FRF has two bandwidths, the dynamic performance of the controller is mainly dominated by the lower one. In other words, to improve the dynamic performance, the controller should be tuned in such a way, that the lower one of the two bandwidths is maximized.

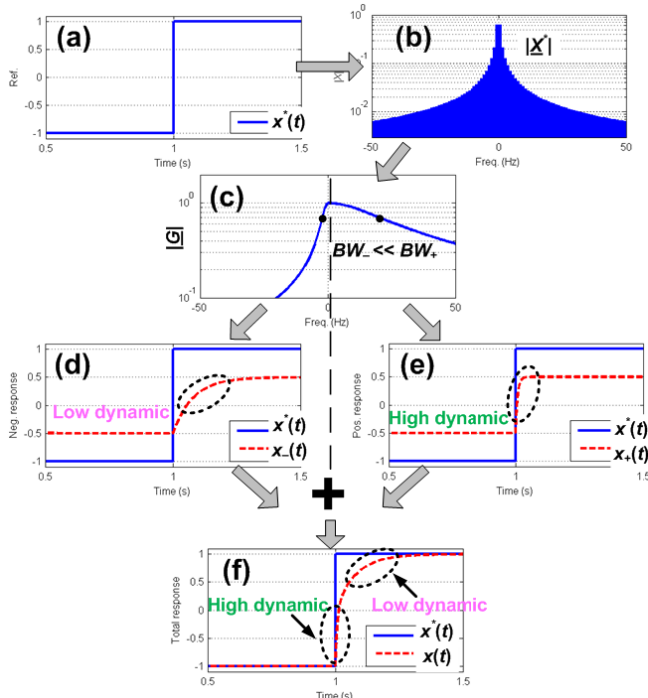


Fig. 19 Dynamic performance with an unbalanced double-sided bandwidth: (a) input reference signal; (b) spectrum distribution of input signal; (c) Transfer function $G(s)$ with unbalanced bandwidth; (d) d-axis response contributed by the negative part of $G(s)$; (e) d-axis response contributed by the positive part of $G(s)$; (f) observed d-axis response waveform

D. Symmetry of the double-sided FRF vs. cross-coupling

It is well known that the (dq) cross-coupling effect of the PI controls can be intrinsically decoupled by enabling a SFB

[6]. However, the concept “coupling” refers to a MIMO system that is valid for e.g. matrix modeling methods. By using the complex function modeling, the control loop is a SISO. In such systems, the “cross-coupling” effect can be interpreted as the symmetry of the transfer function: it is discussed that the spectrum of a step signal is symmetric around dc (i.e. equal amplitude distribution but conjugated phase angles). Similarly, if the transfer function is also symmetric, the real part of the response signal will be added and the imaginary parts are compensated, i.e. it leads to a perfect decoupling between the real and the imaginary axis. In real controllers there are always certain asymmetries in the open-loop and closed-loop transfer functions, so that a perfect decoupling is not achievable.

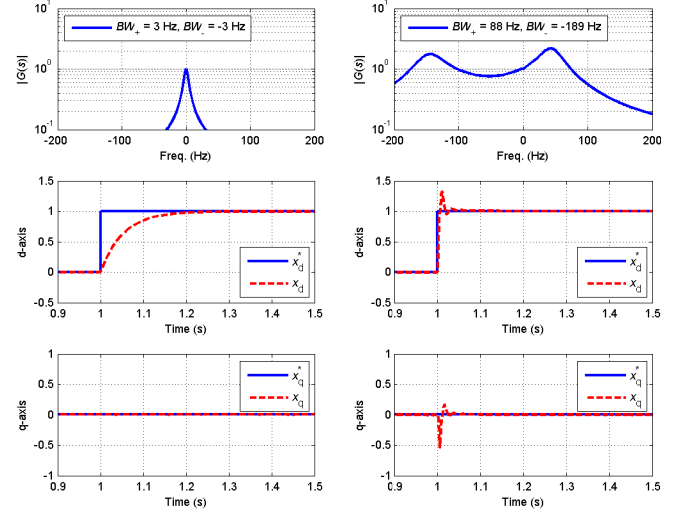


Fig. 20 Symmetry of FRF vs. cross-coupling effect: left: response of a symmetric $G(s)$ with low bandwidths; right: response of an asymmetric $G(s)$ with high bandwidths

It should be also highlighted that the “symmetry” of the transfer function (i.e. a good decoupling) does not generally lead to a higher dynamic performance of the controller. As shown in Fig. 20, a controller may have a perfect decoupling with low dynamics, or a strong cross-coupling with very high dynamics. Although it is widely accepted that a SFB leads to a significantly improved dynamic, it should be noted that this is due to the increased bandwidth (see TABLE I), which is just a side effect of the “decoupling”.

E. Duality between dq-frame and $\alpha\beta$ -frame controls

From the examples in Fig. 6 it can be found that there is no intrinsic performance difference whether a controller is implemented in the dq-frame or in the $\alpha\beta$ -frame, since the FRFs remain unchanged except for a frequency shift of ω_0 . E.g. to regulate the negative sequence in the dq-frame, a PI controller only needs to add an additional resonant term $1/(s+j2\omega_0)$ that is actually a “frequency-shifted” integrator, see (7).

On the other hand, the FRF of a PR controller may be symmetrical in the $\alpha\beta$ -frame (depending also on e.g. the phase compensation setups), but not in the dq-frame. In other words, the cross-coupling effect still exists between the d and

q axis in a PR controller, which can also be observed from a PI controller. At the end, the frame reference signals (e.g. the phase angle or the rotating frequency) are the key factors that determine the performance difference between the dq-frame and the $\alpha\beta$ -frame controllers. Detailed explanations will be given in a future paper.

$$\begin{aligned} PR &= K_p + \frac{K_I}{s_{\alpha\beta} - j\omega_0} + \frac{K_I}{s_{\alpha\beta} + j\omega_0} \\ &= K_p + \underbrace{\frac{K_I}{s_{dq}}}_{\text{standard PI}} + \underbrace{\frac{K_I}{s_{dq} + j2\omega_0}}_{\text{additional pole}} \end{aligned} \quad (7)$$

V. CONCLUSIONS

In this paper, a modeling and analysis concept for the control loop designs for high power converters with extremely low sampling ratios is presented. Using the complex transfer function modeling, the intrinsic MIMO dq-frame controller becomes a SISO. By extending the frequency axis to the negative part, the control system can be characterized with FRFs in an intuitive form. The precision of the analytical FRF models are well proven via circuit simulations in SABER.

It is shown in this paper that for three-phase current controllers, two phase margins at both the positive and the negative frequency ends must be considered for the stability analysis. Similarly, two bandwidths at both the positive and the negative frequency ends must be considered, since the lower one dominates the overall dynamic performance of the control loop. In other words, it is more efficient to maximize the lower one of the two bandwidths. By using the complex transfer function modeling, the de-coupling effect of the controller is interpreted as the symmetry of the double-sided FRF. Generally, there is no intrinsic benefit or drawback to implement a current controller in the dq or $\alpha\beta$ -frame, since their transfer functions can be identically expressed, with only a fundamental frequency shift.

Regarding applications for high power converters with extremely low sampling ratios, it is discovered that the phase margins and the bandwidths are the general challenges. Typical phase compensation for the PI controls leads to an insufficient phase margin at the negative frequency, and leads to an insufficient phase margin at the positive frequency for the PI+SFB controls. A proper phase compensation term can effectively balance the margins at both ends by shifting the phase of the FRF. However, due to an asymmetric transfer function, it seems to be difficult for the standard PI controls to obtain a sufficient total phase margin, and its two bandwidths are strongly unsymmetrical. These lead to a poor dynamic performance and a strong cross-coupling effect of the PI control. In contrast, the PI+SFB has significant advantages at low sampling ratios in terms of the phase margin, dynamic performance and the d-q axis decoupling.

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